Measurement Uncertainty of STP02-FDT-TM3

FC03-1-TM3-MU-240219

1. Model equation

a. Test excavation holes/shafts dimension or "bell-out" dimension (L')

L' = r * S

Where

r	:	Measured value on UES profiles by a calibrated caliper
S	:	Scaling factor (to convert r into L')
L'	:	Converted range value

σr/σL = S

σS/σL = r

input data

locked cell

Factors affecting the uncertainty of each input

		Value	Coverage factor	
caliper	calibration uncertainty,mm			
	resolution (read to 0.01mm)			
	tolerance of between operators reading			(only 1 operator)
	correction			
Scaling factor	calibration uncertainty, m/mm			
	rounding (read to 4 significant figure)			
	S, m/mm	X-X, mm:	Y-Y, mm	Sample

Symbol	Input	Factor	a	Distribution	ci (sensitivity coeff.)	u(xi)*ci	ci*u(xi)^2	γ	ci*u(xi)^4/γ
S	Calibration factor	Calibration	#DIV/0!	normal	0	#DIV/0!	#DIV/0!	10000	#DIV/0!
		Rounding	#DIV/0!	rectangular	0	#DIV/0!	#DIV/0!	10000	#DIV/0!
r	Caliper	Calibration	#DIV/0!	normal	0	#DIV/0!	#DIV/0!	10000	#DIV/0!
		Resolution	#DIV/0!	rectangular	0	#DIV/0!	#DIV/0!	10000	#DIV/0!
		Taking reading	#DIV/0!	rectangular	0	#DIV/0!	#DIV/0!	10000	#DIV/0!
		Correction	#DIV/0!	rectangular	0	#DIV/0!	#DIV/0!	10000	#DIV/0!
						sum	#DIV/0!		#DIV/0!
						uc	#DIV/0!		

						Veff	#DIV/0!		
						t value	2		
						U95	#DIV/0!	m	

Result L' (X-X')= S*X-X' 0.000 m ± #DIV/0! m

Symbol	Input	Factor	a	Distribution	ci (sensitivity coeff.)	u(xi)*ci	ci*u(xi)^2	γ	ci*u(xi)^4/γ
S	Calibration factor	Calibration	#DIV/0!	normal	0	#DIV/0!	#DIV/0!	10000	#DIV/0!
		Rounding	#DIV/0!	rectangular	0	#DIV/0!	#DIV/0!	10000	#DIV/0!
r	Caliper	Calibration	#DIV/0!	normal	0	#DIV/0!	#DIV/0!	10000	#DIV/0!
		Resolution	#DIV/0!	rectangular	0	#DIV/0!	#DIV/0!	10000	#DIV/0!
		Taking reading	#DIV/0!	rectangular	0	#DIV/0!	#DIV/0!	10000	#DIV/0!
		Correction	#DIV/0!	rectangular	0	#DIV/0!	#DIV/0!	10000	#DIV/0!
						sum	#DIV/0!		#DIV/0!
						uc	#DIV/0!		
						Veff	#DIV/0!		
						t value	2		
						U95	#DIV/0!	m	

Result L' (X-X')= S*X-X' 0.000 m ± #DIV/0! m

b. Verticality/ inclination

$$V = \frac{DF}{(A - B)} \qquad I = \frac{\text{(Sum of horizontal shift) * calibration factor}}{\text{DF (length of pile)}}$$

Where

A	reading measured from UES profiles near the top region of the first received signals (in m)
B	reading measured from UES profiles near the bottom of the first received signals (but before a “bell-out” region, if any) (
DF	Depth difference between the depths at which the values of readings A and B are measured on the UES profiles (in m)

Measured Horizontal Shift (X - dir.) = X, mm 0.00 Axx Bxx

Measured Horizontal Shift (Y - dir.) = Y, mm 0.00 Ayy Byy

Sum of Shift (X-X' & Y-Y'),mm $= \sqrt{X^2 + Y^2}$ 0

Sum of shift, m		0
DF (pile length), m		

σX/σI=	X*S/(√ X2 + Y2)*DF	#DIV/0!
σY/σI =	Y*S/(√ X2 + Y2)*DF	#DIV/0!
σDF/σI =	-sum/DF^2 = -√ X2 + Y2 /DF^2	#DIV/0!
σS/σI =	(√ X2 + Y2)/DF	#DIV/0!

Factors affecting the uncertainty of each input		Value	Coverage factor
DF	Tolerance of verification, m		
	Uncertainty of tape, m		
	correction		
Caliper	calibration uncertainty, m		
(assuming)	resolution (read to 0.01mm), m		
	tolerance of between operators reading, m		
	correction, m		
Calibration factor	calibration uncertainty, m/mm		
	rounding (read to 4 significant figure)		

Symbol	Input	Factor	a	Distribution	ci (sensitivity coeff.)	u(xi)*ci	ci*u(xi)^2	γ	ci*u(xi)^4/γ
DF	Verification	Tolerance	#DIV/0!	normal	#DIV/0!	#DIV/0!	#DIV/0!	10000	#DIV/0!
	Tape	Calibration	#DIV/0!	normal	#DIV/0!	#DIV/0!	#DIV/0!	10000	#DIV/0!
		correction	#DIV/0!	rectangular	#DIV/0!	#DIV/0!	#DIV/0!	10000	#DIV/0!
S	Calibration factor	Calibration	#DIV/0!	normal	#DIV/0!	#DIV/0!	#DIV/0!	10000	#DIV/0!
		Rounding	#DIV/0!	rectangular	#DIV/0!	#DIV/0!	#DIV/0!	10000	#DIV/0!
Axx	Calipar	Calibration	#DIV/0!	normal	#DIV/0!	#DIV/0!	#DIV/0!	10000	#DIV/0!
		Resolution	#DIV/0!	rectangular	#DIV/0!	#DIV/0!	#DIV/0!	10000	#DIV/0!
		Tolerance	#DIV/0!	rectangular	#DIV/0!	#DIV/0!	#DIV/0!	10000	#DIV/0!
		Correction	0.0000000	rectangular	#DIV/0!	#DIV/0!	#DIV/0!	10000	#DIV/0!
Bxx	Calipar	Calibration	#DIV/0!	normal	#DIV/0!	#DIV/0!	#DIV/0!	10000	#DIV/0!
		Resolution	#DIV/0!	rectangular	#DIV/0!	#DIV/0!	#DIV/0!	10000	#DIV/0!
		Tolerance	#DIV/0!	rectangular	#DIV/0!	#DIV/0!	#DIV/0!	10000	#DIV/0!

		Correction	0.0000000	rectangular	#DIV/0!	#DIV/0!	#DIV/0!	10000	#DIV/0!
Byy	Calipar	Calibration	#DIV/0!	normal	0.0000000	#DIV/0!	#DIV/0!	10000	#DIV/0!
		Resolution	#DIV/0!	rectangular	0.0000000	#DIV/0!	#DIV/0!	10000	#DIV/0!
		Tolerance	#DIV/0!	rectangular	0.0000000	#DIV/0!	#DIV/0!	10000	#DIV/0!
		Correction	0.0000000	rectangular	0.0000000	0	0.00E+00	10000	0.000E+00
Y'	Calipar	Calibration	#DIV/0!	normal	#DIV/0!	#DIV/0!	#DIV/0!	10000	#DIV/0!
		Resolution	#DIV/0!	rectangular	#DIV/0!	#DIV/0!	#DIV/0!	10000	#DIV/0!
		Tolerance	#DIV/0!	rectangular	#DIV/0!	#DIV/0!	#DIV/0!	10000	#DIV/0!
		Correction	0.0000000	rectangular	#DIV/0!	#DIV/0!	#DIV/0!	10000	#DIV/0!
						sum	#DIV/0!		#DIV/0!
						uc	#DIV/0!		
						Veff	#DIV/0!		
						t value		2	
						U95	#DIV/0!		

Result (I) =	#DIV/0!		±	#DIV/0!	
Inclination , 1:	#DIV/0!				
	range		#DIV/0!	#DIV/0!	
Inclination , 1:			#DIV/0!	#DIV/0!	
Conclusion:	X-X'	0.000	±	#DIV/0!	m
	Y-Y'	0.000	±	#DIV/0!	m
	Inclination, (I)	#DIV/0!	±	#DIV/0!	
Inclination, 1:	range	#DIV/0!	#DIV/0!	#DIV/0!	

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Date:		Date:	

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